

Indian Statistical Institute
Second Semester Examinations: 2024-25

Course Name: M. Math, 1st year
Subject Name : Topology II
Maximum Marks: 60, Duration: Three hours
Date: 8.5.2025, 2:30 PM-5:30 PM

- Answer as many questions as you can.
- Maximum marks is 60.
- You may use any results proved in class. Any other results (including those in homework problem sets) require proof.

1. Let $X = S^1 \times S^1$ and $A = \{p_1, \dots, p_k\} \subset X$. Compute the homology groups $H_n(X, A)$. 10
2. Let $f : S^{2n} \rightarrow S^{2n}$ be a map. Show that there exists $x \in S^{2n}$ with $f(x) = x$ or $f(x) = -x$. 10
3. Let X be a path connected CW complex such that $\pi_1(X)$ is a finite group. Prove that for any based map $X \rightarrow S^1 \times S^1$, the induced map $\tilde{H}_*(X) \rightarrow \tilde{H}_*(S^1 \times S^1)$ is 0. 10
4. Let X be a CW complex with finitely many cells, $C(X)$ be the cone of X , and $p \in C(X)$ be the cone point. Compute $H_*(C(X), C(X) \setminus \{p\})$. 10
5. Consider the map q defined as the following composite
$$q : \mathbb{R}P^{2n+1} = S^{2n+1}/C_2 \rightarrow \mathbb{C}P^n = S^{2n+1}/S^1,$$
defined by identifying the group C_2 as the subgroup of S^1 given by $\{\pm 1\}$.
 - a) Let $\mathbb{R}P^{n-1} \rightarrow \mathbb{R}P^n$ and $\mathbb{C}P^{n-1} \rightarrow \mathbb{C}P^n$ be the inclusions that are given by the formula $[x_0, \dots, x_{n-1}] \mapsto [x_0, \dots, x_{n-1}, 0]$. Prove that q restricts to a homeomorphism $\mathbb{R}P^{2n} \setminus \mathbb{R}P^{2n-1} \rightarrow \mathbb{C}P^n \setminus \mathbb{C}P^{n-1}$. 5
 - b) Prove that $q_* : H_{2n}(\mathbb{R}P^{2n+1}; \mathbb{Z}/2) \rightarrow H_{2n}(\mathbb{C}P^n; \mathbb{Z}/2)$ is an isomorphism. 10
 - c) Let $f : \mathbb{R}P^{2n+1} \rightarrow \mathbb{C}P^n$ be a map homotopic to q . Prove that f is surjective. 10

